

Semester Project

Analyzing Binary Search vs Linear Search Using Calculus and Python

Introduction

Searching algorithms are fundamental to computer science and play an important role in data retrieval and processing. Among the most commonly used searching algorithms are Linear Search and Binary Search, each with different time complexities and performance characteristics. This project aims to explore the mathematical behavior of both algorithms, derive analytical expressions representing their performance, and compare them using calculus and Python. You are required to perform the following tasks in the given order.

Project Tasks

1. Introduction and Objectives

- Provide a brief introduction, objectives, and solution methodology.
- Provide flowcharts or block diagrams of your solution methodology or Python codes.

2. Mathematical Modeling of Search Performance

- Consider an array of size n . Derive the expected number of comparisons required for:

Linear Search

$$L(n) = \frac{n + 1}{2}$$

Binary Search

$$B(n) = \log_2(n)$$

- Explain the significance of each term in the expressions.
- Implement the symbolic expressions $L(n)$ and $B(n)$ in PYTHON.

3. Understanding Growth Rates and Complexity

- Compare the growth rates of Linear Search and Binary Search using calculus by computing the following derivatives:

$$\begin{aligned} & \frac{d}{dn} L(n) \\ & \frac{d}{dn} B(n) \end{aligned}$$

- Interpret the meaning of the derivatives in terms of algorithmic efficiency.
- Verify your results using PYTHON symbolic differentiation.

4. Optimization and Performance Thresholds

a. PYTHON Task

- i. Determine the value of n at which Linear Search becomes significantly slower than Binary Search (e.g., where $L(n) > 10B(n)$).
- ii. Solve this inequality using Python.

b. Manual (by-hand) Task

- i. Manually solve the inequality or approximate the result.
- ii. Compare your manual findings with PYTHON output.

5. Visualization and Analysis

a. PYTHON Task

- i. Plot both functions $L(n)$ and $B(n)$ on the same graph.
- ii. Plot their derivatives on a separate graph.
- iii. Analyze how the performance gap grows as n increases.

b. Manual (by-hand) Task

- i. Sketch approximate graphs for $L(n)$ and $B(n)$.
- ii. Mark the point where Binary Search becomes significantly more efficient.

6. Comparative Analysis

a. PYTHON Task

- i. Generate a table showing the number of comparisons for various values of n (e.g., 10, 100, 1000, 10000).
- ii. Compare actual values of both search algorithms.

b. Manual (by-hand) Task

- i. Discuss differences and similarities between manual and Python-generated results.
- ii. Provide interpretation:
 - a) Why does Binary Search scale better?
 - b) Why does Linear Search become inefficient for large n ?

c) Conclusions

- a. Conclude your project in this section.

d) Contributions

- a. Clearly specify the contributions of each group member.

Report Requirements

Students are required to submit a complete report of the project prepared in MS Word in their own words. A report is written in third person, i.e., I, We, Us, are not used for writing the report. The headings of the report should be the same as the heading and subheadings of the tasks given above. Overall, the project marks are divided into following criteria:

Sr. No.	Criteria	Marks
1.	Objectives and Introduction: Objectives and introduction of the problem. In this section briefly introduce the problem and the methodology that will be adopted by you to solve the problem. Also include flowcharts of the solution methodology or the programs.	10
2.	Analytical Solution: A step-by-step analytical solution (by-hand solution) for all the tasks as specified in the 'Project Tasks'. Clearly state the assumptions and values that you use for the solution.	20
3.	PYTHON Code: Complete and well commented PYTHON codes. Each task's code must include the explanation of the commands, functions, and toolboxes used.	15
4.	Solution and Results: Present detailed results and discussion for each task. Do not just paste the graphs or screenshot of the command window. Compare your by-hand and PYTHON solutions, and present physical interpretation of your results and graphs.	20
5.	Conclusions: In this heading, include conclusions related to this project. The conclusion section stands independently from the report and gives the reader a comprehensive idea of the project; thus, the conclusion section should briefly explain the problem, solution methodology, results, and analysis. The conclusion section is not very large and typically consists of 1-2 paragraphs. The conclusions section can also include bullet points.	5
6.	Contribution: In this heading clearly state the contribution of each group member. Generic statements such as 'each group member contributed equally' are not acceptable answers. In this section include difficulties that you faced during this assignment and how you overcame those difficulties.	5

Each project task should be documented under a separate heading and sub-heading in the same order as given in the 'Project Tasks'. Each page should be numbered. The report should be written in the Times New Roman typeface only. The size of the font should be 12. The size of first and second level of headings should be 14 bold, and 12 bold, respectively. The alignment of the report should be justified, while pictures and tables should be center aligned with relevant captions. The option to align the text left, right, center, and justify can be found under paragraph options on *Home* tab. Line and paragraph spacing should be set as 1.5. Optimally utilize the available space on each page, do not leave blank space on a page unnecessarily.

Guidelines

This project is an open-ended problem designed to demonstrate the application of Calculus in real life in general and Computer Science in particular. The open-ended nature of the problem means that this problem can be solved in more than one way using various techniques and methodologies, some of these techniques have been covered in this course. You are free to adopt any technique and solution methodology to solve this problem. Solution techniques and methodologies that are not part of the course outline can also be used to solve the problem. However, you are required to take approval of such a solution technique before starting the project. You may have to do extensive research to completely solve the problem. If you have any confusion, you can discuss your query via email. Project guidelines are summarized below:

- This is a group project and carries 75 marks.
- A group can have a maximum of 3 students. One of the aims of this project is to enable students to work effectively in a team. **Therefore, this project cannot be done individually.** The project can be done in pairs.
- Plagiarized work (from internet, AI, or fellow students) will result in zero marks. Similarity index of less than 18% and AI content less than 15% is allowed.
- Deadline for complete project submission on [google classroom](#) ([one MS Word file and one pdf of the same Word file including all the codes and by-hand solutions](#)) is **30 Nov 2025 before 11:59pm**. Do not submit your project in a .zip or .rar format. You can submit additional files, however, the single PDF and MS Word file must also include all the information such as codes and figures.
- Name of your project report file must be as per following format: **ID1_ID2_ID3_MT1003_Project_Section.**
- Do not submit your project via email, it will not be considered.
- Late submissions will not be considered.

